MILITARY SPECIFICATION

TESTER, TYPE MA-2 RECEPTION COUPLING BREAK-AWAY FORCE, PORTABLE

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a portable tester used for checking the break-away force required to disengage the type MA-2 reception coupling.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Box, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation, Methods of
DOD-D-1000	Drawing, Engineering and Associated List

STANDARDS

FEDERAL

FED-STD-151	Metal, Test Methods

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, Texas 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4920

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MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-889	Dissimilar Metals
MS24536	Tester, Type MA-2 Reception Coupling Break-Away Force, Portable

* (Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.1.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

* AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 Standard Practice for Commercial Packaging.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

* 2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

* 3.1 First Article. When specified, a sample shall be subjected to first article inspection (see 4.6 and 6.3).

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials.

3.3.1 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be encountered in storage or normal service.

3.3.1.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

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3.4 Design and construction. The design and construction of the tester shall conform to MS24536. The tester shall be as light as possible consistent with good rugged construction and shall incorporate a calibrated pressure gage reading directly in pounds force from 0 to 1,000 pounds. The tester shall be hydraulically actuated by the turning of a hand wheel. There shall be no leakage of hydraulic fluid.

3.4.1 Maintenance. The tester shall have a minimum number of parts consistent with reliability. Its design shall, where practical, permit easy assembly, disassembly, location of trouble sources, and maintenance with tools and equipment normally available commercially, by service maintenance personnel with a minimum of training.

3.5 Performance. The tester shall be capable of operation by one man and perform its functions as required herein.

3.5.1 Calibration accuracy. The tester shall maintain an accuracy within plus or minus 5 percent of the correct reading in the range of 25 to 800 pounds when tested in accordance with 4.5.2.

3.5.2 Temperature range. The tester shall operate satisfactorily from -65°F to ±130°F. The torque required to actuate the tester shall not be more than 40 pound-inches. The torque required to reset the tester shall not be more than 10 pound-inches.

3.5.3 Endurance. The tester shall operate satisfactorily throughout the endurance test with no leakage or other malfunction.

3.5.4 Corrosion resistance. The tester shall withstand 50 immersion and heat cycles to a temperature of 130° plus or minus 5°F as specified in 4.5.6.1 and salt spray for a period of 50 hours as specified in 4.5.6.2

3.6 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of DOD-D-1000 shall govern changes in the manufacturer's part numbers.

3.7 Dimensions. The dimensions ahall be as shown on MS24536.

3.8 Weight. The weight of the tester shall not exceed 10 pounds.

3.9 Identification of product. The tester, and all parts, shall be marked for identification in accordance with MIL-STD-130.

3.10 Workmanship. All details of workmanship shall be in accordance with the best commercial practice covering this type of equipment.

* 3.11 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless

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disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the tester shall be classified as follows:

a. Acceptance tests See 4.3

b. First Article Inspection See 4.6

4.3 Acceptance tests. Acceptance tests shall consist of individual tests.

4.3.1 Individual tests. Each tester shall be subjected to the following tests as described under 4.4.

a. Examination of product

b. Calibration tests

4.4 Test conditions. The test conditions shall be as specified herein for each separate test.

4.5 Test methods.

4.5.1 Examination of product. Each tester shall be examined for conformance to this specification with regard to materials, workmanship, dimensions, and markings. The tester shall be mounted in a vertical position, with the dummy nozzle down and the unit supported by the nozzle withdrawal spider, and operated through one cycle to check for binding of other malfunction.

4.5.1.1 Cycle. A test cycle shall consist of the tester being operated, by turning the hand crank, so that the dummy nozzle travels from the extended or normal engagement position to the retracted position, a minimum travel of 0.75 inch, and is returned to the extended position.

4.5.2 Calibration accuracy. The calibration accuracy shall be checked by using dead weights of 25, 100, 300, 400, 600, and 800 pounds. Each weight shall be accurate within plus or minus one-fourth pound. The tester shall be mounted as specified in 4.5.1 and the weight shall be hung by means of a cable attached to the fitting inside the dummy nozzle. The weight should be supported so that there will be no load exerted on the cable when the tester is in the extended position. The tester shall be cycled once with each of the calibrating weights. As the dummy nozzle is retracted the weight shall be lifted from its support and the gage must read within plus or minus 5 percent of the dead weight.

4.5.3 Low temperature. The tester shall be maintained at a temperature of -65°F for a period of 12 hours. The tester shall then be cycled once with each of the calibrating weights listed in 4.5.2. There shall be no evidence of leakage of the hydraulic fluid or other malfunction. The gage reading must be within the limits specified in 4.4.2.

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4.5.4 High temperature. The tester shall be maintained at a temperature of 130° for a period of 12 hours. The tester shall then be operated once with each of the calibration weights listed in 4.5.2. There shall be no evidence of leakage of hyraulic fluid or other malfunction. The gage reading must be within the limits specified in 4.5.2.

4.5.5 Endurance. The tester shall be operated through 1,000 cycles at room temperature using the 800-pound calibration weight.

4.5.6 Corrosion resistance. Corrosion resistance shall include both accelerated and salt spray tests.

4.5.6.1 Accelerated corrosion. The tester shall be immersed in a solution consisting of 2.5 percent sodium chloride in distilled water. After immersion, the tester shall be drained and heated to a temperature of 130°F plus or minus 5°F for a period of 1 hour. This immersion and heating cycle shall be repeated 50 times. The tester shall not be actuated at any time during the above test. Immediately after completing the immersion and heat cycles, the tester shall be washed with warm water to remove all salt accumulations, after which the tester shall be dried and actuated as in normal service. The force required to operate the handle shall not exceed that specified in 3.5.2. There shall be no evidence of deterioration that might adversely affect subsequent operation.

4.5.6.2 Salt spray. The tester shall be subjected to a salt spray in accordance with FED-STD-151 for a period of 50 hours. At the end of the test period, the salt deposits shall be removed with warm water. The tester shall be dried and actuated as in normal service. The force required to operate the handle shall not exceed that specified in 3.5.2. Any evidence of corrosion that might adversely affect the operation of the tester shall be cause for rejection.

4.6 First Article Inspection. The first article tests shall consist of all the tests specified under 4.5.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or commercial, as specified (see 6.2).

5.1.1 Level A. The tester shall be preserved and packaged one each, method IC2 of MIL-P-116 and shall be individually overboxed in a fiberboard unit container conforming to PPP-B-636, class domestic.

5.1.2 Commercial. Testers shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be Level A, B, or Commercial, as specified (See 6.2).

5.2.1 Level A. Testers packaged as specified in 5.1 shall be packed in wood-cleated plywood shipping containers conforming to PPP-B-601, overseas type. Gross weight of the shipping container shall not exceed 200 pounds. Container closure and strapping shall be in accordance with the appendix to PPP-B-601.

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5.2.2 Level B. Testers packaged as specified in 5.1 shall be packed in wood-cleated plywood shipping containers conforming to PPP-B-601, domestic type gross weight of the shipping container shall not exceed 200 pounds. Container closure and strapping shall be in accordance with the appendix to PPP-B-601.

* 5.2.3 Commercial. Testers preserved as specified in 5.1 shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special markings required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with the requirements of MIL-STD-129. The nomenclature shall be:

"TESTER, TYPE MA-2 RECEPTION COUPLING, BREAK-AWAY FORCE, PORTABLE"

6. NOTES

6.1 Intended use. The portable tester covered by this specification is intended for use in checking the toggle settings to determine the break-away force of the type MA-2 reception coupling. The tester can be used to check the toggle settings either individually or combined while either installed or removed from the aircraft.

* 6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable levels of preservation, packaging and packing (see 5.1 and 5.2).

6.3 First Article. When a first article inspection is required, the item will be tested and should be a first article sample or it may be a standard production item from the contractor's current inventory. The first article should consist of two testers. First articles should be identified with the manufacturer's own part number and the contracting officer should include specific instructions for examinations, tests and approval of the first article.

6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

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c. Reason/Rationale for Recommendation:

6. REMARKS

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